ESCROWMIND

AI Freelance Escrow Agent

WORK WITH TRUST. PAY WITH CONFIDENCE.

Project submission documentation

Live application: escrowmind-web.onrender.com/workspace

Source code: github.com/divyanshuwu/Escrow-Mind

Demo video: youtube.com/watch?v=nDPI1ey6nIM

Prepared for hackathon judges and organizers

1. Project Overview

EscrowMind is an AI-assisted workflow for freelance projects. It is designed to reduce uncertainty between clients and freelancers by connecting project requirements, milestone-based work, deliverable review, and payment status in one place.

Freelancers may worry about being paid after completing work, while clients may worry about paying before receiving satisfactory deliverables. EscrowMind aims to make expectations and progress more visible to both parties.

2. Problem and Proposed Solution

Problem: Freelance agreements can be vague, deliverables may be disputed, and payment timing can be unclear. Informal communication makes it difficult to track what was agreed and what remains outstanding.

Solution: EscrowMind structures projects into milestones, associates each milestone with expected deliverables and payment amounts, and supports a review process before eligible payment actions are taken.

3. Core User Journey

Step	Action	Purpose
1	Client enters project description, budget, and expectations.	Create a shared project brief.
2	AI-assisted workflow structures the brief into milestones.	Clarify scope and deliverables.
3	Client proceeds through the configured PayPal payment flow.	Connect payment to the project.
4	Freelancer submits milestone deliverables.	Record work for review.
5	AI-assisted review compares work with agreed requirements.	Support consistent evaluation.
6	Workflow shows next step, such as client review or eligible payment action.	Keep status transparent.

4. AI Components

AI Contract Agent: Helps turn a natural-language project description into structured scope, milestones, deliverables, and acceptance expectations.

AI Verification Agent: Helps compare submitted work against the requirements recorded for a milestone and return a verification result or feedback.

Human oversight: AI output should support, not replace, fair review. Client review and clear dispute-handling rules remain important, especially for subjective creative work.

5. Milestone and Payment Workflow

The workspace example includes three design milestones: Initial Concepts & Logo Sketches ($50), Selected Concept Refinement ($70), and Final Assets & Vector Source Delivery ($80), representing a $200 example project. These example values do not imply that a real payment has occurred.

Milestone-based payments can make expectations clearer by associating each payment portion with a defined output. Actual authorization, capture, refund, and payout behavior depends on the PayPal products and configuration used.

6. PayPal Integration and Important Scope

EscrowMind is intended to use PayPal payment infrastructure in its payment workflow. The project materials reference authorization and capture concepts, with payout handling as part of the broader design.

Important: PayPal checkout or payment authorization is not automatically the same as a legally or technically segregated escrow account. Exact behavior depends on the APIs used, account eligibility, region, and implementation. During judging, clearly identify sandbox versus live behavior and whether each action is real or simulated.

7. Technology and Architecture

The project materials describe a web application using Next.js and TypeScript, PostgreSQL through Supabase for data management, GPT-4o for AI-assisted contract and deliverable analysis, and PayPal for payment operations. Confirm exact versions and enabled integrations against the submitted repository.

CLIENT / FREELANCER		
EscrowMind Web App **Next.js + TypeScript interface** **PROJECT WORKFLOW** **Brief • milestones • submissions • status**		
AI SERVICES	**DATA LAYER**	**PAYMENT PROVIDER**
GPT-4o Contract assistance & deliverable review	Supabase / PostgreSQL Project and workflow data	PayPal APIs Configured payment actions

8. Security and Responsible Use

API secrets and payment credentials must remain server-side and must never be committed to source control or included in the submission. Use environment variables and sandbox credentials for testing. Validate payment events server-side, verify webhook signatures where applicable, and make payment operations idempotent to reduce duplicate processing.

AI verification can be imperfect, especially for subjective deliverables. The product should explain review criteria, preserve a human review path, and avoid relying solely on opaque model scores. Minimize and protect personal and payment data.

9. Challenges and Learnings

Important challenges in this problem space include translating informal project descriptions into testable acceptance criteria, reviewing varied file types and creative work, handling disagreements fairly, and aligning payment states with project states. The project demonstrates why AI assistance must be paired with clear user consent, payment safeguards, and human review.

10. Demo and Evaluation Guide

A useful judge walkthrough is: create or open a sample project, inspect its milestones, review payment status, submit a deliverable, inspect AI review output, and show the next eligible workflow step. Clearly label any mocked or sandbox-only behavior.

Live application: Open EscrowMind Workspace

Source code: View GitHub Repository

11. Future Scope

Potential improvements include richer dispute workflows, configurable acceptance criteria, audit logs, multi-currency support, additional deliverable formats, stronger webhook reconciliation, fraud and anomaly signals, accessibility improvements, and clearer explanations of AI review outcomes.

12. Project Demonstration Video

The following YouTube video is included as supplementary project media for hackathon judges and organizers.

WATCH ESCROWMIND PROJECT DEMO

Video URL: https://youtu.be/nDPI1ey6nIM

Platform: YouTube

Submission note: Ensure the video is Public or Unlisted, has no password, and is under the hackathon's three-minute limit.

This document is a project overview. Implementation details, live payment status, and production readiness should be verified directly in the submitted build.